

Review

The effect of gas transport on the isotope signature of methane in wetlands

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Abstract

This paper reviews the ^{13}C isotopic fractionations of the three main modes of CH_4 transport from wetlands: diffusion, ebullition and via emergent aquatic plants. Aquatic plants employ either diffusion or a combination of convective bulk flow in day-light and diffusion at night. Ebullition and diffusion across the air–water interface result in no to little isotopic fractionation. However, diffusion through air when conducted by plant lacunae or aerenchyma results in significantly greater rates of transport for $^{12}\text{CH}_4$ relative to $^{13}\text{CH}_4$. This difference causes the backup of ^{13}C -enriched CH_4 in plant lacunae, which partly but not totally compensates for the isotopic fractionation out of the plant. It is also suggested that the preferential plant transport of $^{12}\text{CH}_4$ causes ^{13}C enrichment of methane in the rhizosphere but it is difficult to separate the effects of rhizospheric methane oxidation from this process. Convective bulk flow transport via plants results in little isotopic fractionation. However, the pattern of fractionation differs significantly in convective plants from day to night as diffusion is more important in darkness.

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1. Use of stable isotopes

Stable isotopes are extremely useful for diagnosing a wide variety of processes associated with determining methane production mechanisms (Whiticar et al., 1986; Whiticar, 1993, 1999; Hornibrook et al., 1997, 2000; Lansdown et al., 1992), the extent of methane oxidation (Barker and Fritz, 1981; Coleman et al., 1981) or tracking sources of methane (Martens et al., 1991). Applications vary across a wide variety of scales. On a small scale, shifts in methane $\delta^{13}\text{C}$ can be used to calculate fractional oxidation in the transport of methane across landfill and wetlands soils (Borjesson et al., 2001; Happell et al., 1994). On a global scale, the isotopic composition of methane emitted from natural and anthropogenic sources is used for constraining the global methane budget (Stevens and Engelkemeir, 1988; Tyler, 1989, 1991; Reeburgh, 2003). In many cases, methane stable isotopic signatures are affected by gas transport. To attribute variations in the isotopic signature of methane to the correct process requires a thorough understanding of the factors that affect the transport of methane from one reservoir to another. As stated by Tyler et al. (1997), "In our own research, an understanding of the role of the different emission methods proved to be a key."

2. Three modes of methane transport from wetlands

Methane is transported from anoxic sediments and flooded soils by three main pathways: molecular diffusion, bubble ebullition and via emergent plants (Schutz et al., 1991, Fig. 1). The relative importance of these three pathways varies according to organic loading, seasonal variation in temperature and the type and density of aquatic vegetation.

Molecular diffusion is the random movement of molecules down a concentration gradient, from higher to lower concentration. Rates of diffusion are regulated by the concentration gradient, the diffusion coefficient and, in sediments and soils, the porosity. Diffusion follows Fick's first law (Berner, 1980):

$$J = \phi D dC/dz, \quad (1)$$

where J is the diffusive flux (units of mass per unit area per time), D is the diffusion coefficient (units of $\text{length}^2/\text{time}$), ϕ is the porosity (unitless and defined as the volume of water/volume of whole sediment or soil) and dC/dz is the concentration gradient or the change in concentration (C , $\mu\text{mol}/\text{cm}^3$), as a function of depth or distance (z , cm).

Molecular diffusion is important in the transport of methane transport from open water (Happell and Chanton, 1993; Happell et al., 1995), from sediments, particularly in winter (Chanton and Martens, 1988; Sorrell and Boon, 1992) or from peat bogs where the ground cover is mainly sphagnum (Chasar et al., 2000a,b; Fechner and Hemond, 1992).

Bubble ebullition occurs because methane is only sparingly soluble in water (Yamamoto et al., 1976). Ebullition transports methane directly to the atmosphere, bypassing the mediating effects of an oxygenated water column or plant rhizome. In the process of sedimentary bubble formation, dissolved methane and other gases equilibrate with the bubble reservoir in the sediment and are subsequently stripped from solution as bubbles are released (Kipphut and Martens, 1982; Chanton et al., 1989; Martens and Chanton, 1989). Bubble N_2 content can serve as an indicator of the importance of differing gas transport mechanisms (e.g., ebullition versus plant transport, Boon and Sorrell, 1991; Sorrell and Boon, 1992) or may be indicative of rates of ebullition from sediments (Chanton et al., 1989; Martinova, 1998). As increasing volumes of methane bubble from the sediments, they strip out N_2 , a gas that is not produced within the sediments. Thus, bubbles with low N_2 concentration indicate high rates of bubble stripping, while bubbles with high N_2 concentration indicate that little ebullition has removed N_2 from the sediment.

The volume of free gas in the sediment can be as great as three times the dissolved gas inventory (Fechner-Levy and Hemond, 1996); volumes of gas of up to 14 $\text{liters} \cdot \text{m}^{-2}$ have been observed (Chanton et al., 1989). Bubble release can be triggered by pressure differences driven by variations in current or tide (Martens and Klump, 1980; Chanton et al., 1989; Scranton et al., 1993), wind events (Keller

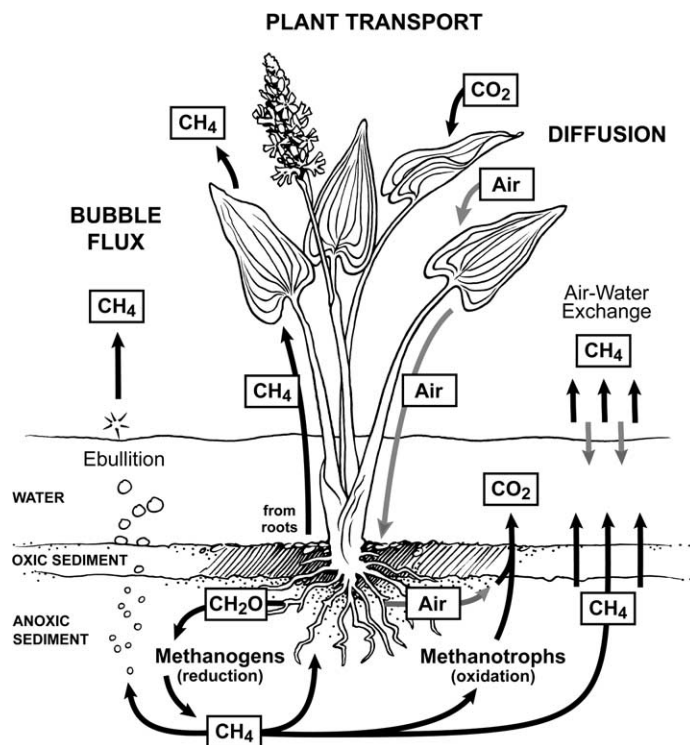


Fig. 1. The three mechanisms of gas transport from wetlands are by emergent aquatic plants, by ebullition and by molecular diffusion (Schutz et al., 1991).

and Stallard, 1994), falling hydrostatic pressure associated with variation in river flood stage (Bartlett et al., 1988; Devol et al., 1988; Smith et al., 2000) or atmospheric pressure (Mattson and Likens, 1990; Fechner-Levy and Hemond, 1996; Casper et al., 2000; Glaser et al., 2004). Ebullition is episodic and thus it is very difficult to quantify its importance. Consequently, methane ebullition may be much greater than is generally thought (Romero et al., 1996; Glaser et al., 2004). In lakes, ebullition can dominate methane emissions (Casper et al., 2000). In lakes and reservoirs, methane bubbles originate from the upper portion of the sediment (Joyce and Jewell, 2003) where methane production rates are greatest. Ebullition is more frequent in shallow waters (<5 m) where bubbles are released by bottom currents and surface winds (Keller and Stallard, 1994). The rates vary seasonally, being lower in winter and greater in warmer months (Chanton and Martens, 1988; Martens and Chanton, 1989; Boon and Sorrell, 1995).

Aquatic plants conduct considerable quantities of methane as natural wetlands and artificial wetlands (rice fields) are two of the largest sources of methane in the atmosphere (Cicerone and Oremland, 1988; Reeburgh, 2003). Many emergent aquatic macrophytes have large interior open spaces, termed lacunae or aerenchyma, within their culms, stems or petioles (Dacey, 1981a,b). These plants release methane from the sediments while transporting oxygen to support the respiration of their roots

and rhizomes, which often live in organic-rich anoxic sediments and soils (Schutz et al., 1991; Chanton and Dacey, 1991; Shannon and White, 1994; Shannon et al., 1996; King et al., 1998). In comparisons of vegetated areas with adjacent non-vegetated areas, it has been found that plants generally transport 10 times the amount of CH_4 relative to the non-vegetated interface (Whiting et al., 1991; Whiting and Chanton, 1992; Happell et al., 1993).

Ventilation of CH_4 by plants can affect gas transport mode by reducing methane partial pressures within the flooded soil or sediment in which they are rooted. Where plants are present, pore water methane concentrations are roughly 50% lower due to plant ventilation of the soil and this in turn reduces the methane concentration gradient out of the soil, reducing diffusive flux (Wilson et al., 1989; Chanton et al., 1989; Chanton and Dacey, 1991). Another effect of reducing pore water CH_4 concentrations is to reduce CH_4 bubble concentration and total bubble volume, both of which reduce ebullition rates. Thus, if vascular emergent macrophyte plants inhabit a wetland, plant transport will be the primary mechanism of CH_4 transport from the wetland (Van der Nat and Middelburg, 1998). In vegetated areas, CH_4 transport is dominated by plants while in non-vegetated ones CH_4 transport is dominated by ebullition and diffusion (Kaki et al., 2001; Tyler et al., 1997; Boon and Sorrell, 1991). Macrophyte population is controlled

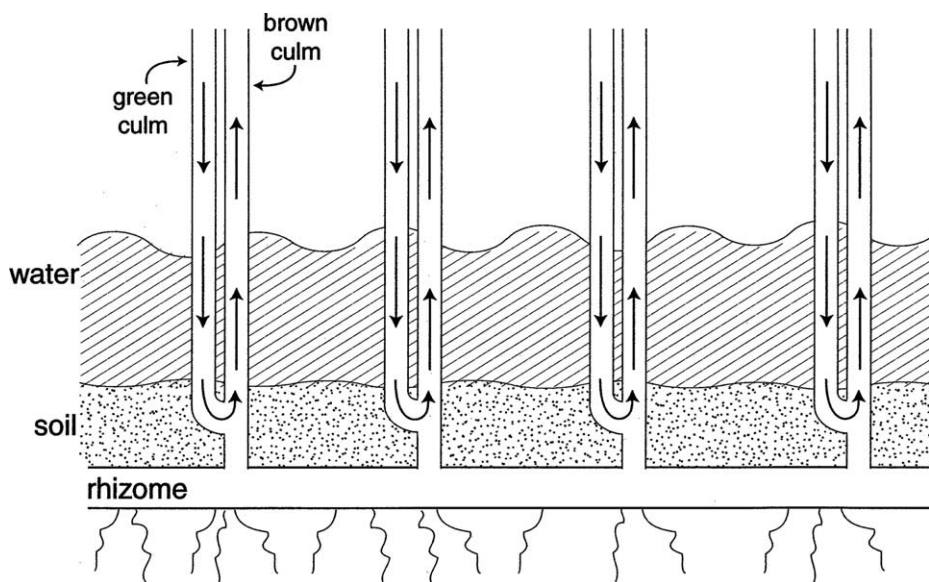


Fig. 2. In *Phragmites* daylight pressurization (Arkebauer et al., 2001) drives air flow down green culms and out paired brown culms (Chanton et al., 2002). At night, the pressure is relaxed and molecular diffusion dominates methane transport out of the culms. Methane and CO_2 concentration reflect gas transport mode. At night, under a diffuse regime, their concentrations are greater. During daylight, pressurized convective through-flow sweeps CH_4 and CO_2 from the culms. Rates of 10 mL/min have been observed.

by shading and water depth. Thus, diffusive transport is important in methane emission from the shaded floors of swamp forests (Happell and Chanton, 1993).

Aquatic macrophytes may employ passive (diffusive) gas transport systems or active gas flow systems (Dacey and Klug, 1979; Dacey, 1981a,b; Armstrong and Armstrong, 1990, 1991; Grosse et al., 1991, 1996; Schutz et al., 1991; Van der Nat et al., 1998; Whiting and Chanton, 1996; Van Bodegom et al., 2001; Colmer, 2003). Passive (diffusive) transport is the sole gas transport mode found in shallow water plants such as *Carex rostrata*, *Oryza sativa* and *Peltandra virginica* (Chanton and Dacey, 1991; Van Bodegom et al., 2001).

In deeper water, plants that employ diffusive transport are at a competitive disadvantage relative to plants that use convective through-flow ventilation. Diffusion is less adequate for spanning longer distances between the atmosphere and the roots, so active modes of gas transport give certain plants an advantage in deeper water (Grosse et al., 1991). Sorrell et al. (1997) found that for *Eleocharis sphacelata* diffusion is inadequate for satisfying oxygen demands in plants growing in more than 10 cm of water and that convection is essential in the natural habitat of the species. Convective inflows in this plant increased with greater water depth (Sorrell and Tanner, 2000).

Active transport is the convective through-flow of gases from areas of high pressure to low pressure. The pressure gradient can be created by either thermal transpiration (Dacey, 1981a,b; Grosse et al., 1991; Grosse, 1996) or it can be induced from a humidity differential across the leaf boundary (Armstrong and Armstrong,

1990, 1991; Armstrong et al., 1992, 1996a,b; Brix et al., 1992, 1996; Sorrell and Boon, 1994; Sorrell et al., 1997; Arkebauer et al., 2001). Pressurization of young green leaves causes through-flow to porous dead or senescent leaves (Dacey, 1981a,b; Armstrong et al., 1992; Bendix et al., 1994). Pressurization sets up a gas flow through the plant culms or petioles as illustrated in Fig. 2 for *Phragmites australis*. Gas flow rates of 10 mL/min were observed by Chanton et al. (2002).

Plants that utilize convective through-flow transport include *Typha latifolia*, *Typha angustifolia*, *Nymphaea odorata*, *Nuphar luteum*, *Nelumbo nucifera*, *Nymphoides peltata*, *Eleocharis sphacelata* and *Phragmites australis*. Convective through-flow transport generally occurs in the daylight (Dacey, 1981a,b; Grosse and Mevi-Schutz, 1987; Mevi-Schutz and Grosse, 1988; Armstrong and Armstrong, 1991; Bendix et al., 1994; Brix et al., 1992, 1996; Tornbjerg et al., 1994; Arkebauer et al., 2001). At night, when pressure gradients within the plants are absent, molecular diffusion is the primary transport mechanism (references above). However, small increases in CH_4 flux on windy nights have been contributed to Venturi-induced convection in *Phragmites australis* (Armstrong et al., 1992, 1996a; Brix et al., 1992; Kim et al., 1998).

It appears that the diurnal pattern of CH_4 emission from aquatic macrophytes is strongly influenced by the mode of gas transport (Kaki et al., 2001). In aquatic plants having only diffusive transport, only small variations in flux rate have been observed and attributed to changes in soil and water temperature (Whiting and Chanton, 1996; Van der Nat et al., 1998). However, a

dramatic short burst increase in CH₄ emission rate during daylight has been observed in *Typha*, a convective transport plant (Chanton et al., 1993; Whiting and Chanton, 1996; Yavitt and Knapp, 1995). The *Typha* emission pattern is characterized by a sharp peak in mid-morning and emissions are not closely correlated with PAR intensity (Chanton et al., 1993; Whiting and Chanton, 1996). In *Typha*, daylight switches on convective through-flow and the elevated concentration of lacunal CH₄ is flushed, resulting in an emission peak. Following this peak, emission rate is similar to night time rates, suggesting that CH₄ flux is limited by gas transport across the root–shoot boundary as suggested by Kelker and Chanton (1997). In *Phragmites*, the diel pattern of emissions appears to vary, being more like *Typha* in the report of Van der Nat et al. (1998) but with a more sustained peak following PAR intensity in the report by Kim et al. (1998).

Lacunal CH₄ concentration is affected by gas transport mechanism (Whiting and Chanton, 1996; Van der Nat et al., 1998). In *Typha*, the lacunal CH₄ concentrations were high in the pre-dawn and early morning hours and decreased rapidly at mid-morning, staying low throughout the day. At night, CH₄ again accumulated in the culms. The morning-time rapid decrease for lacunal CH₄ concentration is associated with the initiation of convective through-flow transport. Lacunal CH₄ accumulates in the evening as convective transport becomes less active. Similar behavior in lacunal gas concentrations was observed in *Phragmites* (Chanton et al., 2002; Van der Nat et al., 1998). In *Peltandra* and *Scirpus*, both passive transport plants, the lacunal concentration remained relatively constant over a 24 h period (Whiting and Chanton, 1996; Van der Nat et al., 1998).

3. Stable isotope terminology

There are two useful natural “tags” for methane, the ¹³CH₄/¹²CH₄ ratio and the CH₃D/CH₄ ratio. Carbon-13 is about 1.1% abundant while D (²H) is only 0.01% abundant (Hoefs, 1997). Stable isotope notation describes the ratio of the heavy isotope to the light isotope in the sample of interest ($R_{\text{sam}} = {}^{13}\text{C}/{}^{12}\text{C}$ sample) relative to a known standard ratio, R_{std} , which is PDB for carbon and SMOW for hydrogen

$$\delta\text{‰} = [(R_{\text{sam}}/R_{\text{std}}) - 1]1000. \quad (2)$$

Isotopic fractionation for the reaction $A \rightarrow B$, is expressed as

$$\alpha = R_A/R_B, \quad (3)$$

where R_A and R_B represent ¹³C/¹²C of A and B, respectively. From Eq. (2) above,

$$\alpha = [(\delta_A/1000) + 1]/[(\delta_B/1000) + 1]. \quad (4)$$

Another useful term for expressing isotopic fractionations is the term Δ , where

$$\Delta = |\delta_A - \delta_B| \approx (\alpha - 1)1000 \quad (\text{Hoefs, 1997}). \quad (5)$$

This paper discusses mainly fractionation of $\delta^{13}\text{C}$ rather than δD . To measure isotopic transport effects only trace amounts of methane can be captured. It has only recently been possible to measure trace samples for δD of CH₄, with the development of continuous flow hydrogen isotopic analysis (Rice et al., 2001; Chidthaisong et al., 2002; Yamada et al., 2003).

4. Isotopic fractionation by diffusion

Isotopic fractionation caused by transport processes is important for characterizing the isotopic value of the methane flux from particular environments and for revealing subtleties in the transport processes. Additionally, patterns of isotopic fractionation are used to determine methane oxidation (Happell et al., 1994; Kelley et al., 1995; Lombardi et al., 1997; Tyler et al., 1997; Bilek et al., 1999; Popp et al., 2000) and methane production (Whiticar et al., 1986; Whiticar, 1999; Lansdown et al., 1992; Kelley et al., 1992, 1995; Martens et al., 1992; Hornibrook et al., 1997, 2000; Chanton et al., 2005). The fractionations associated with gas transport can obscure and confuse interpretation of these latter two biological processes.

4.1. Fractionation caused by differences in diffusion coefficients

In the transport of methane from wetlands, diffusive transport can occur from the surface of a water body or soil or as methane diffuses through the gas within plants. Fractionation during diffusion can be caused either by differences in the diffusion coefficients of the two species, or by an isotopic gradient along the diffusive path (Chanton et al., 1987). As discussed below, the second mechanism can compensate for the first, resulting in less fractionation.

In air, the diffusion of the lighter ¹²CH₄ is more rapid than the diffusion of heavier ¹³CH₄. Thus, $D_{12} > D_{13}$, where D_{12} represents the diffusion coefficient for ¹²CH₄ and D_{13} represents the diffusion coefficient for ¹³CH₄.

Similar to calculations made for the gaseous diffusion of CO₂ in air through the boundary layer and into the stomata of a leaf, where a fractionation of 4.4‰ is predicted (Farquhar et al., 1982; Ehleringer et al., 2002), the diffusion coefficient (D) of a gas in air is inversely proportional to the square root of the reduced masses of the gas of interest and of air (Mason and Marrero, 1970) as given below:

$$D \text{ is proportional to } [(mMm)/(mM + mA)]^{-1/2}, \quad (6)$$

where mM is the molecular weight of methane and mA is the molecular weight of air (28.8, the weighted average of O₂ + N₂). We can solve for the ratio of the diffusivities of ¹²CH₄ and ¹³CH₄ as shown below:

$$D_{12}/D_{13} = [(m_{12}mA)/(m_{12} + mA)]^{-1/2} / [(m_{13}mA)/(m_{13} + mA)]^{-1/2} \quad (7)$$

and obtain 1.0195, where D_{12} and m_{12} are the diffusivity and mass of $^{12}\text{CH}_4$, and D_{13} and m_{13} are the diffusivity and mass of $^{13}\text{CH}_4$. The same fractionation ought to apply for CH_3D because this molecule has the same mass as $^{13}\text{CH}_4$. Thus the diffusion of methane in air may cause an isotopic fractionation on the order of 19‰ because CH_4 containing the lighter isotope diffuses through air more rapidly than the CH_4 containing the heavy isotope.

In the diffusion of methane across the air–water interface, the fractionation is much smaller. The rate limiting step in the transfer of a gas out of water occurs in the diffusive aqueous sub-layer just under the air–water interface. The difference in diffusion coefficients as a function of mass is much smaller in water than in air.

The diffusive flux of gases across the water–air interface is generally modeled with the equation:

$$F = K(C_w - C_o), \quad (8)$$

where F is the flux, K the transfer coefficient, C_w is the measured concentration of the gas dissolved in water and C_o is the calculated concentration of the dissolved gas in equilibrium with the atmospheric gas concentration (for more detail see MacIntyre et al., 1995). The mass-transfer coefficient, K , is a function of water temperature, wind speed, water turbulence and the diffusion coefficient of the gas. The kinetic isotope fractionation factor (α_k) is equal to the ratio of the exchange coefficients:

$$\alpha_k = (K_{12}/K_{13}). \quad (9)$$

This factor has been experimentally determined in the laboratory for methane diffusing out of still, pure water to be 1.0008 (Knox et al., 1992; Happell et al., 1995), indicating isotopic fractionation on the order of 1‰, with the lighter isotope crossing the air–water interface faster than the heavy isotope. Under conditions of greater turbulence in the water, this fractionation would presumably be less as there is less fractionation associated with turbulent transfer relative to diffusion. Under stagnant conditions which are often found in wetlands and swamps, organic surface films and slicks can reduce gas exchange rate relative to that measured in pure standing water and increase isotopic fractionation to 2–3‰ (Happell et al., 1995). When diffusion of methane out of water occurs, the remaining dissolved CH_4 becomes ^{13}C enriched.

4.2. Compensation for diffusive fractionation caused by isotopic gradients

Diffusive transport is a very important mode of methane transport from the flooded floor of hardwood swamp forests, such as those found in the southeastern USA (Happell and Chanton, 1993). Happell et al. (1994) presented a conceptual model for the transport

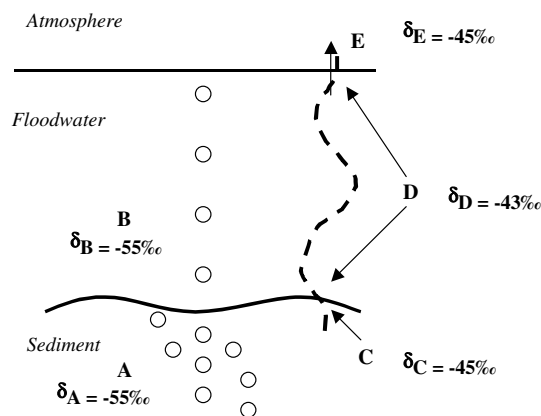


Fig. 3. Processes that affect CH_4 as it is transported from flooded soils and sediments to the atmosphere (Happell et al., 1994). (A) CH_4 bubbles in the main production zone, slightly below the zone of oxidation. The isotopic composition of the CH_4 in this zone is labeled $\delta\text{A} = -55\text{‰}$. (B) CH_4 in bubbles is not subject to oxidation because it travels through the oxidation zone too rapidly, $\delta\text{B} = \delta\text{A} = -55\text{‰}$. (C) CH_4 oxidation occurs at the sediment–floodwater interface. δC is the isotopic composition of methane diffusing out of the oxidation zone ($\delta^{13}\text{C} = -45\text{‰}$, which is enriched relative to δA). This CH_4 is input to the bottom of the water column. δD is the isotopic composition of the methane within the floodwater. δE is the isotopic composition of the flux of methane across the air–water interface. If one assumes that oxidation occurs predominantly at the sediment–water interface, that oxidation is nil within the floodwater and that the system is at steady state, then the isotopic composition of CH_4 entering the floodwater at C must be identical to the isotopic composition of methane emitted from the floodwater at E, $\delta\text{C} = \delta\text{E} = -45\text{‰}$. However, a slight isotopic fractionation was observed at the E boundary, the lighter isotope being emitted more rapidly (Happell et al., 1995). This fractionation was compensated for by the increasing isotopic enrichment in the floodwater, (D). So δD is ^{13}C enriched (-43‰) relative to δC and δE . The shift in isotopic value between δE relative to δA is indicative of the extent of methane oxidation at the sediment water interface. This conceptual model was consistent with oxidation determinations using picolinic acid (Happell et al., 1994).

of methane from the flooded forest floor (Fig. 3). Transport was dominated by ebullition and diffusion. Diffusive transport of methane across the air–water interface was associated with a fractionation of 2–3‰. The model assumed that methane oxidation occurred only at the flooded soil–water interface. Lack of methane consumption within the floodwater meant that at steady state, input and output to the floodwater must be balanced. Therefore, the quantity and isotopic composition of methane diffusing across the soil–water interface into the floodwater was equivalent to the quantity and isotopic composition of methane diffusing from the surface of the water into the atmosphere. But if there was isotopic discrimination at the upper boundary, how could this be?

The isotopic discrimination at the upper boundary (the air–water interface) caused the buildup of CH₄ containing the heavy isotopes of methane within the floodwater. This relative increase in the concentration gradient in ¹³CH₄ and CH₃D eventually compensated for the discrimination against the CH₄ containing the heavy isotopes in their transport across the air–water interface. Preferential transport of the lighter isotopes of methane out of the floodwaters across the air–water interface was caused by the larger diffusion coefficient of the CH₄ containing lighter isotopes. However, the effect of this discrimination caused the heavier isotope to back up within the floodwater and thus compensated for this discrimination by causing a steeper gradient in the heavy isotope of methane. The floodwater could be thought of as a transport pipeline. What goes in the pipe at one end must be the same as what comes out of the pipe at the other end if there is no consumption or production within the pipe and if the system is at steady state. A second example of this phenomenon is presented below.

4.3. Fractionation when diffusive supply limits consumption

Diffusion of methane into sub aerial soils is a different case. Methane is consumed within forest soils by methanotrophic bacteria that have affinity for the very low concentration in the atmosphere, i.e., 1.8 ppmv (Born et al., 1990; Keller et al., 1990; Crill, 1991). Methane is supplied to these soil bacteria by diffusion from the atmosphere into the soil. Often the bacteria are associated with the soil mineral layer that is several centimeters below more or-

ganic-rich layers (Crill, 1991). In an elegant study, Tyler et al. (1994a) observed that the rate of microbial consumption of methane was limited by the diffusion of methane into the soil. Since all of the methane that diffused into the forest soil was consumed, they noted that the differences in the diffusion coefficients of methane controlled the total isotopic fractionation associated with forest soil methane uptake, not the enzymatic fractionation of the oxidizing bacteria. The bacteria quantitatively consumed all of the methane that reached them without discrimination. The fractionation occurred in the transport to them and the isotopic composition of what reached them was controlled by the relative differences in the diffusion coefficients ($D_{12}/D_{13} = 1.0195$), as described above. However, while the fractionation by diffusive transport is relatively well defined, variation in fractionation factors for methane oxidation by microbial process remains relatively undefined (Reeburgh, 1996) and may vary with CH₄ concentration and temperature (Reeburgh et al., 1997). Reeburgh et al. (1997) used three approaches to estimate the biological fractionation of methane uptake in forest soils and determined that the overall isotopic fractionation was controlled by a combination of transport processes and microbial discrimination.

5. Fractionation by ebullition

Chanton and Martens (1988) and Martens and Chanton (1989) clearly established that the gas composition and methane $\delta^{13}\text{C}$ value of released bubbles were

Table 1

The $\delta^{13}\text{C}$ of CH₄ naturally released by bubble ebullition versus $\delta^{13}\text{C}$ of CH₄ within the sediments released by stirring (Numbers in parentheses indicate the standard deviation of replicate samples if $n = 3$ or range of values if $n = 2$)

Site	$\delta^{13}\text{C}$ of CH ₄	<i>n</i>
Cape Lookout Bight, NC, USA 22 September, 1984		
Bubbles	−57.5 (0.2)	3
Reservoir	−58.6 (0.2)	2
Cape Lookout Bight, NC, USA 11 September, 1985		
Bubbles	−58.6 (0.4)	2
Reservoir	−56.6 (0.1)	2
Cape Lookout Bight, NC, USA 5 August, 1986		
Bubbles	−58.2 (0.2)	3
Reservoir	−57.4 (0.4)	4
Old Mill Creek, DEL, USA 19 June, 1986		
Bubbles	−60.5 (0.1)	2
Reservoir	−61.6 (0.3)	2
White Oak River, NC, USA 21 July, 1986		
Bubbles	−67.3 (0.5)	2
Reservoir	−66.4 (0.1)	2
White Oak River, NC, USA 13 August, 1986		
Bubbles	−66.7 (0.1)	3
Reservoir	−67.2 (0.1)	3

Data from Chanton and Martens (1988); Martens et al., 1986, and Martens and Chanton (1989).

not different from the reservoir of bubble gas held within the sediment (Table 1). Thus, bubbles stirred from the sediment are a representative sample of CH₄ released due to a natural perturbation. Methane transported via bubble ebullition may appear relatively ¹³C depleted as the gas is transported directly and rapidly from reducing environments to the atmosphere, bypassing the ¹³C-enriching sites of methanotrophy (cf. Happell et al., 1994).

6. Isotopic fractionation associated with transport by aquatic plants

6.1. Gas transport modes

There are three modes of gas transport by which methane is transported out of a plant: (1) effusion, also called free-molecular or Knudsen flow, which can be described as the passage of gas in the presence of a pressure differential, through a partition with holes which are small relative to the mean free path of a gas molecule in air, (2) viscous, bulk or convective flow, in which the gas acts as a continuum driven by a pressure gradient, or (3) diffusion, described by Fick's law, in which the gas responds to the influence of a composition or partial pressure gradient in the absence of a total pressure gradient (Dacey, 1987; Mason and Evans, 1969; Mason and Marrero, 1970).

Of these three mechanisms, only the second, bulk or convective flow, does not result in a molecular weight dependent fractionation. This kind of flow takes place in the transfer from the inlet system to the ionization chamber in the isotope ratio mass spectrometer. Effusion is the principle behind the separation of ²³⁵U from ²³⁸U in the gas phase as hexafluorides. This technique was used by Ramsey to separate He from N₂ through unglazed clay piping (Kennard, 1938). Dacey (1987) was able to characterize pore sizes in living leaves of *Nelumbo* using fractionation of N₂ and Ar in effusion experiments. Schröder (1989) demonstrated fractionation of ethane and propane during gas transport in the black alder in order to estimate pore sizes in lenticel tissues. Harden and Chanton (1994) used ethane and propane fractionation to demonstrate diffusive fractionation in gas transport from *Pontederia cordata* (Pickerel weed) and *Sagittaria lancifolia* (Bull tongue). Van der Nat et al. (1998) used flux variations under differing chamber atmospheres to deduce gas transport mechanisms.

Effusive flux (J) is inversely proportional to the molecular weight of a gas according to Graham's law (Mason and Marrero, 1970):

$$J_{12}/J_{13} \text{ is proportional to } De_{12}/De_{13} = (M_{13}/M_{12})^{1/2}, \quad (10)$$

where De is the effusive diffusion coefficient and M is the molecular weight of the gas (Mason and Evans, 1969; Mason and Marrero, 1970). Effusive flux could lead to a maximum isotopic fractionation for methane of 30.7‰ ($\alpha = 1.0307$, $\Delta = 30.7\text{‰}$; Mason and Evans, 1969; Mason and Marrero, 1970). However, depending upon pore size of the holes within the partition, effusive flux can occur in conjunction with bulk flow, thus reducing the amount of isotopic fractionation.

6.2. Patterns of isotopic fractionation

In the mid 1980s, a number of research groups characterized the $\delta^{13}\text{C}$ value of atmospheric methane sources (Stevens and Engelkemeir, 1988; Cicerone and Oremland, 1988; Tyler, 1989, 1991) to constrain estimates of the relative quantities of the inputs that were fueling the rapid (1% per year) increase in atmospheric methane (Cicerone and Oremland, 1988). It was hypothesized that methane transported out of the sediments by emergent aquatic plants would be very enriched in ¹³C and D due to rhizospheric methane oxidation (Chanton et al., 1988). Significant rates of methane oxidation occur when material from the rhizosphere is incubated with methane under O₂ (King, 1990, 1994; Chanton et al., 1992b; Gerard and Chanton, 1993; Moore et al., 1994; Moore and Dalva, 1997). Methanotrophic bacteria in the rhizosphere were envisioned as being a gauntlet that the methane would have to cross as it diffused from anaerobic production sites through plants to the atmosphere.

The first samples collected to evaluate the isotopic composition of methane transported by plants were from the lacunal gases of plants collected in the Florida (USA) Everglades (Chanton et al., 1988). These lacunal gases were enriched in ¹³CH₄ relative to methane within the flooded soil. A hypothesis was proposed that the isotopic composition of lacunal CH₄ was representative of the CH₄ emitted from the plants and that the isotopic composition of CH₄ released from wetlands was enriched by 5–10‰ relative to below ground CH₄.

The first comparisons of the $\delta^{13}\text{C}$ of CH₄ emitted from a plant compared with the lacunal CH₄ $\delta^{13}\text{C}$ within the plant showed this hypothesis to be incorrect (Chanton et al., 1992a,b). In these two studies, the isotopic composition of all the relevant pools of methane was determined. Methane enriched in ¹³C was present within the lacunal spaces of *Peltandra virginica*, *Arctophila fulva*, and *Carex rostrata* and was also present within the standing floodwaters, but the methane emitted from plants was similar to, or even ¹³C depleted relative to, sediment methane (Table 2). Clearly, plant transport did not convey ¹³C-enriched methane to the atmosphere. It appeared that isotopic fractionation on the order of 12–18‰ occurred at the plant atmosphere boundary due to mass dependent gas transport,

Table 2

The $\delta^{13}\text{C}$ value for methane sampled from sediment (from bubbles or pore water equilibrators) from plant lacunae, and emitted from vegetation and the difference in $\delta^{13}\text{C}$ between samples

Plant type	Day/Night	Sediment	Plant	Emitted	$\Delta 1$	$\Delta 2$	$\Delta 3$	ref ^c
Diffusive plants ^a								
<i>Peltandra virginica</i> 1	D	−58	−52	−66	6	−14	−8	1
<i>Peltandra virginica</i> 2	D	−56	−49	−61	7	−12	−5	1
<i>Peltandra virginica</i> 3	D	−57	−50	−64	7	−14	−7	1
<i>Peltandra virginica</i> 4	D	−57	−55	−70	7	−15	−13	1
<i>Oryza sativa</i>	D	−50	−43	−54	7	−11	−4	2
<i>Oryza sativa</i>	D	−51	−42	−56	9	−14	−5	2
<i>Oryza sativa</i>	D	−49	−41	−53	8	−12	−4	2
<i>Oryza sativa</i>	D	−54	−42	−55	12	13	−1	2
<i>Arctophila fulva</i>	D	−55	−46	−59	9	−13	−4	3
<i>Carex rostrata</i>	D	−61	−48	−67	13	−19	−6	3
Convective plants ^b								
<i>Typha domingensis</i> 1	D	−62	−65	−66	−3	−1	−4	4
<i>Typha domingensis</i> 2	D	−63	−65	−66	−2	−1	−4	4
<i>Typha domingensis</i> 1	N	−63	−60	−73	3	−13	−10	4
<i>Typha latifolia</i>	D	−62	−62	−61	0	1	1	4
<i>Typha latifolia</i>	N	−62	−54	−65	8	−11	−3	4
<i>Phragmites</i>	D	−60	−62	−58	−3	4	2	5
<i>Phragmites</i>	N	−60	−55	nd	5	nd	nd	5
<i>Nymphaea odorata</i>	D	−55	−55	−60	0	−5	−5	5

(Note the lesser extent of fractionation in daylight in the convective plants, which increases to resemble the patterns observed in the diffusive plants at night. Day or night is denoted in column 2. $\Delta 1$ = plant – sediment; $\Delta 2$ = emitted – plant, $\Delta 3$ = emitted – sediment. nd = no data).

^a *Peltandra*, *Arctophila*, *Oryza* and *Carex* are plants where gas transport is dominated by diffusion.

^b *Typha*, *Phragmites* and *Nymphaea* have convective through-flow during daylight and diffusion during the night.

^c Reference where 1 = Chanton et al. (1992b); 2 = Tyler et al. (1997); 3 = Chanton et al. (1992a); 4 = Chanton and Whiting (1996); 5 = Chanton et al. (2002).

with $^{12}\text{CH}_4$ being transported from the plant at a faster rate than $^{13}\text{CH}_4$. The build up within the stems of the more slowly transported (heavy isotope-containing) CH_4 species compensated for the fractionation to some degree and resulted in less fractionation in the transport of methane from the sediment to the atmosphere (Table 2). Similar results have been found for rice plants (Table 2), where emitted CH_4 is depleted relative to lacunal CH_4 (Tyler et al., 1997; Chanton et al., 1997; Bilek et al., 1999).

Chanton et al. (1992b) compared sampling of lacunal gas via syringe with squeezing the lacunal gas from petioles in *Peltandra virginica*. The two approaches gave identical results, indicating that syringe sampling was an appropriate mechanism for obtaining lacunal gas samples, at least from plants with large internal spaces. Tyler et al. (1997) argued that removing several of gas from a plant with small internal spaces such as rice might also draw in gases from the rhizosphere. The same considerations apply to *Carex*. In any case, gas sampled from rice and *Carex* appear similar in their fractionation patterns to diffusive plants with larger lacunal spaces (Table 2). This observation indicates that the discrimination effects as gas exits the plant may affect the

rhizosphere and below ground reservoirs also, as discussed further below (Popp et al., 1999).

Continued investigations showed systematic differences (during daylight) between lacunal methane $\delta^{13}\text{C}$ in plants where gas transport was primarily controlled by molecular diffusion and plants that employed pressurized through-flow ventilation (Table 3). Plants that primarily employ diffusion exhibited ^{13}C enriched methane in lacunae while those employing pressurized bulk flow (a non-fractionating process) exhibited CH_4 with $\delta^{13}\text{C}$ values similar to below ground methane (Chanton and Dacey, 1991). Isotopic differences between below ground and lacunal CH_4 serve as an indicator for the dominant gas transport processes in macrophytes (Table 3). Lacunal CH_4 concentration was also greater in plants utilizing diffusion than in those that used pressurized flow (Whiting and Chanton, 1996).

Further investigation of pressurized plants showed that isotopic fractionation varied from day to night in a manner consistent with the transition in gas transport mechanisms from diffusion at night to pressurized convective through-flow during daylight (Chanton and Whiting, 1996; Chanton et al., 2002, Table 2). During the day, methane in lacunae and emitted methane were

Table 3
Isotopic fractionation by emergent macrophytes at sites in the U.S.A

Genus	Site	Stem CH ₄ ‰	Sediment bubble CH ₄ ‰	Δ‰	n
Diffusive plants					
<i>Pontederia</i>	White Oak	−47	−63	16	3
<i>Pontederia</i>	Lake Hall	−41	−53	12	2
<i>Peltandra</i>	Newport	−48	−56	8	4
<i>Peltandra</i>	Everglades	−51	−63	12	2
<i>Sagittaria</i>	Everglades	−60	−64	4	2
<i>Arctophila</i>	Alaska	−46	−55	9	4
<i>Carex</i>	Alaska	−48	−61	13	2
Convective plants					
<i>Nymphaea</i>	Everglades	−74	−64	−10	1
<i>Nymphaea</i>	St. Mark	−64	−62	−2	2
<i>Nymphaea</i>	Lake Hall	−57	−55	−2	2
<i>Typha</i>	Newport	−55	−56	1	3
<i>Typha</i>	White Oak	−62	−63	1	3

Plant stem and sedimentary bubble methane isotopic composition are listed with the difference between them and the number of replicates. The White Oak is a river estuary in North Carolina. Lake Hall and St. Mark swamp are near Tallahassee, Florida. Newport refers to Newport News swamp in Virginia (Chanton and Dacey, 1991).

similar to, or slightly ¹³C depleted relative to, below ground methane. The slight depletion in δ¹³C was hypothesized to be due to the presence of diffusion superimposed upon bulk flow. At night, in pressurized plants, the through-flow system is inactive because this mechanism is light-dependent (Dacey, 1987; Grosse and Mevi-Schutz, 1987; Armstrong and Armstrong, 1990; Arkebauer et al., 2001; Chanton et al., 1993). In darkness, the lacunal methane increases in concentration (Whiting and Chanton, 1996; Chanton et al., 2002) and becomes ¹³C enriched relative to below ground and emitted methane (Tables 2 and 3). Interestingly, the

methane emitted at night from pressurized plants often becomes increasingly ¹³C depleted relative to both methane emitted in daylight and to below ground methane (Table 2).

Large isotopic fractionation was observed in green culms of *Phragmites* at dawn and at dusk during the transition from diffusion to pressurized flow. In green culms at night, methane diffuses from the below ground upwards towards the atmosphere. At daybreak when the sun shines upon the plants, pressurization occurs and downward flow is initiated in green culms, counter to the direction of diffusion. During the transition periods,

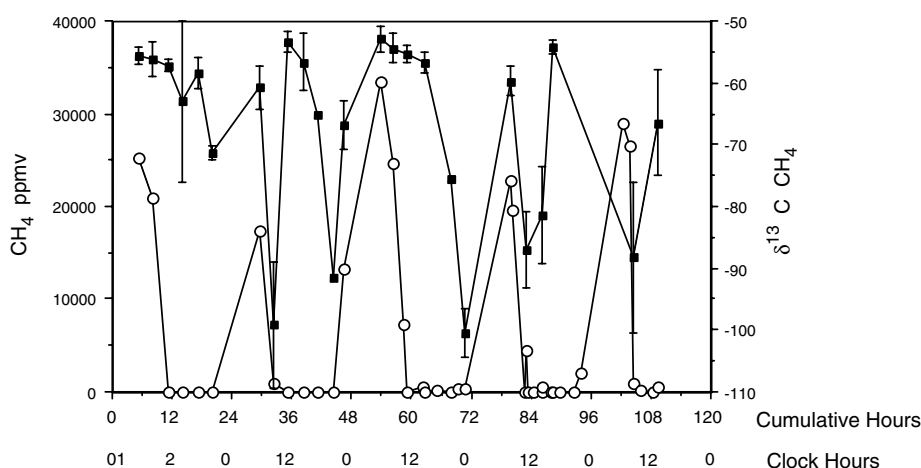


Fig. 4. Green culm lacunal methane concentration (ppmv, open circles) and δ¹³C‰ (filled squares). Extremely depleted isotopic values were observed during times when methane concentration was changing most abruptly during the transition from higher concentration at night when gas transport was dominated by diffusion to lower concentration during the day when convective through-flow flushed accumulated methane from the culms. From Chanton et al., 2002.

when the flow is just beginning or at the end of the day as flow slowly ceases, diffusion and flow have roughly equal strengths in opposite directions in the green culms. At these times CH_4 $\delta^{13}\text{C}$ values as low as -100‰ (Fig. 4) were observed in lacunal methane as the gas diffused upwards in a downwards flow field (Fig. 5; Chanton et al., 2002).

Popp et al. (1999) measured the $\delta^{13}\text{C}$ of CH_4 in a *Carex* fen and characterized emitted methane, rhizospheric methane and the methane below the rhizosphere (Fig. 6). Emitted methane was isotopically similar to methane below the rhizosphere (at 50 cm depth), while methane within the rhizosphere (5–20 cm) was relatively ^{13}C enriched. The ^{13}C enriched methane within the rhizosphere could be due to the preferential transport of $^{12}\text{CH}_4$ from below ground by plants, similar to the effect in the lacunal spaces, but less pronounced due to greater amounts of methane below ground. Additionally, rhizospheric methane oxidation could result in ^{13}C enrichment. Methane at 50 cm is below the bulk of the root zone and free from the influence of oxidation and plant transport. Presumably the $\delta^{13}\text{C}$ at 50 cm represents the value of produced CH_4 , unaffected by oxidation or transport processes. This result, that the isotopic composition of the methane emitted by the wetland was similar to that

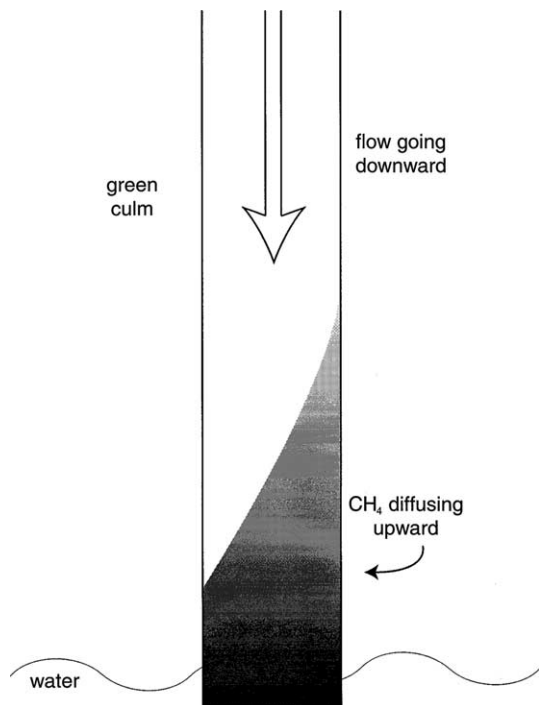


Fig. 5. Extreme isotopic fractionation was observed when CH_4 diffusing upwards encountered downward air flow caused by pressurization. At high flow rates, CH_4 is swept from the green culms, but during transitions times to and from pressurized flow, the two gas mechanisms are of similar strength (Chanton et al., 2002).

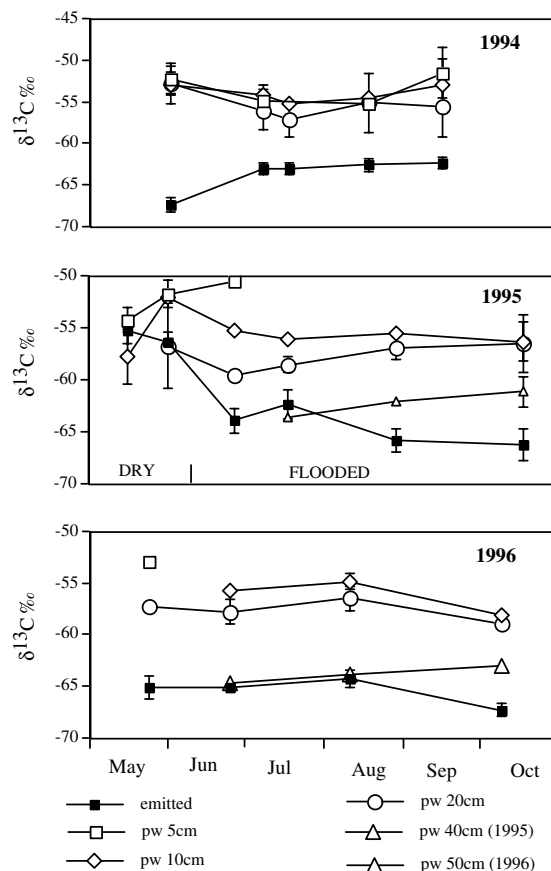


Fig. 6. The $\delta^{13}\text{C}$ values of CH_4 emitted and from various depths within a *Carex* fen in Alberta, Canada. Pore water methane isotopic composition was enriched in $\delta^{13}\text{C}$ relative to emitted CH_4 for all depths except 40–50 cm, the area below the rhizosphere (Popp et al., 1999).

produced there (Fig. 6) seems to indicate that rhizospheric methane oxidation is not an important process.

Several studies have quantified seasonal variation in the $\delta^{13}\text{C}$ and δD values of CH_4 emitted from rice fields (Tyler et al., 1994b; Bergamaschi, 1997; Marik et al., 2002). Isotopically depleted methane has been observed in the main growing season, with more enriched CH_4 at the start and at the end of the season (Tyler et al., 1994b; Marik et al., 2002). Bergamaschi observed ^{13}C depleted methane in the main growing season but δD enriched. These variations have been attributed to variations in methane production (e.g., Fey et al., 2004), oxidation and transport.

7. Future research directions

Currently what is lacking is a comprehensive model (e.g. similar to Kettunen, 2003) that can account for

isotopic fractionations associated with gas transport. With transport effects accounted for, the residual changes in CH_4 isotopic composition could be used to quantify rhizospheric methane oxidation. Preliminary models have been constructed for rice by Chanton et al. (1997), Tyler et al. (1997) and Bilek et al. (1999). Chanton et al. (1997) found that to reproduce the observed isotopic measurements required two and possibly three isotope effects, one associated with methane oxidation and one or two with gas transport. It was difficult to fully constrain the model with so many different possibilities for isotopic fractionation. Tyler et al. (1997) developed a simplified model that was used by Bilek et al. (1999) and Popp et al. (2000). Their model relied upon measurements of the $\delta^{13}\text{C}$ of rhizospheric CH_4 and emitted CH_4 , and corrected the emitted value using the plant transport isotope effect, which was of the order of 11‰. The model produced estimates of the fraction of methane oxidized which were similar to those obtained by other independent techniques. However, it is possible that the model may have been overly simplistic as it included two sinks for ^{13}C depleted CH_4 , oxidation and transport, and did not fully compensate for the build up of residual ^{13}C enriched methane within the rhizosphere and internal plant spaces.

With the advent of continuous flow mass spectrometry for the determination of δD of CH_4 , the problem should be more tractable. Transport-induced isotopic fractionation for the δD of CH_4 should be of similar range to the $\delta^{13}\text{C}$ fractionation, as both are based on mass difference of 1 in 16 (Bergamaschi, 1997). However, evidence suggests that the differences in isotopic fractionation by methanotrophic bacteria is consistently greater for δD of CH_4 than for $\delta^{13}\text{C}$ of CH_4 . Coleman et al. (1981) reported that the δD fractionation was about a factor of 8–10‰ greater during methane oxidation than during the simultaneous fractionation of $\delta^{13}\text{C}$. The results from a number of field studies, compiled by Chanton et al. (2005), suggest less fractionation, of the order of 2–5‰ (Table 4). Further investigation is needed to determine what controls isotopic fraction for $\delta^{13}\text{C}$ and δD during methanotrophy, as well as transport fractionation effects for δD of methane. This research

should aim to produce a model for aquatic plants that simulates transport effects and estimates rhizosphere methane oxidation using both isotopes. CH_4 production effects should also be investigated, for there is no clear consensus on variation of δD with ^{13}C during shifts in production mechanism (Hornibrook et al., 1997; Sugimoto and Wada, 1995; Waldron et al., 1999; Chanton et al., 2005; Fey et al., 2004).

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Table 4

Examples of relative changes in δD of CH_4 relative to $\delta^{13}\text{C}$ of CH_4 caused by methanotrophic activity

Environment	$\Delta\delta\text{D}/\Delta\delta^{13}\text{C}$	Reference
Landfill soil	2.5–3.7	Liptay et al. (1998)
Landfill soil	4	Lubina et al. (1996a,b)
Laboratory incubations	8.5–13.5	Coleman et al. (1981)
Wetlands	5.5	Happell et al. (1994)
Wetlands	2.5	Burke et al. (1988)

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